

Graphing Linear Functions: Domain and Range

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A **linear function** can be written in the form $y = mx + b$, where m is the **slope** and b is the **y -intercept**.

Graph of a Linear Function

The **graph** of a linear function is a straight line.

Graphing a Linear Function

To graph a linear function, create a table of values, plot the ordered pairs, and draw a straight line through the points.

Domain

The **domain** is the set of all possible input values (x -values) of a function.

Range

The **range** is the set of all possible output values (y -values) of a function.

Domain and Range of Linear Functions

For most linear functions of the form $y = mx + b$ where $m \neq 0$, the **domain** is all real numbers and the **range** is all real numbers.

Examples

Example 1

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{3}{2}x - 3$$

Example 2

Graph the linear function and determine its **domain** and **range**.

$$y = -x + 2$$

Example 3

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{2}{3}x - 2$$

Practice Exercises

Practice 1

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{3}{2}x - 3$$

Practice 2

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{1}{2}x - 2$$

Practice 3

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{4}{3}x + 2$$

Practice 4

Graph the linear function and determine its **domain** and **range**.

$$y = -x + 2$$

Practice 5

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{2}{3}x - 2$$

Practice 6

Graph the linear function and determine its **domain** and **range**.

$$y = x - 3$$

Practice 7

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{2}{3}x - 1$$

Practice 8

Graph the linear function and determine its **domain** and **range**.

$$y = -x + 5$$

Activities

Activity 1

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{3}{2}x + 3$$

Activity 2

Graph the linear function and determine its **domain** and **range**.

$$y = -x + 3$$

Activity 3

Graph the linear function and determine its **domain** and **range**.

$$y = x + 1$$

Activity 4

Graph the linear function and determine its **domain** and **range**.

$$y = -\frac{1}{2}x + 1$$

Activity 5

Graph the linear function and determine its **domain** and **range**.

$$y = x - 1$$

Activity 6

Graph the linear function and determine its **domain** and **range**.

$$y = \frac{1}{3}x - 2$$

Activity 7

Graph the linear function and determine its **domain** and **range**.

$$y = x + 2$$

Activity 8

Graph the linear function and determine its **domain** and **range**.

$$y = -x + 4$$

Thank you!